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Render-to-Adapt: Unsupervised Personal Adaptation for Gaze Estimation

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Abstract

Deep learning-based gaze estimation methods tend to suffer from substantial performance drop in real-world scenarios with varying users and environments. To tackle this issue, most recent approaches employ Unsupervised Domain Adaptation (UDA) to bridge the gap between source and target domains. However, this paradigm is misaligned with real-world scenarios, where the system typically needs to adapt to only a single new user. Therefore, this paper advocates a more practical paradigm: Unsupervised Personal Adaptation (UPA), which calibrates a pre-trained model using a few unlabeled images from a single new user. Conventional UDA methods do not guarantee improvements for every user and often yield lower average performance in this setting. To address this problem, we propose Render-to-Adapt (R2A), a self-supervised framework specifically designed for the UPA task. Given a pretrained gaze model, R2A utilizes a gaze-conditioned renderer to synthesize new images based on the model's gaze predictions, and enforces eye-region consistency as a label-free signal to enhance personalized gaze estimation. We evaluate R2A on a re-designed cross-dataset personal adaptation benchmark. Experimental results show that R2A consistently improves performance across all individuals and significantly outperforms existing SOTA methods.

1. Introduction

Gaze estimation is increasingly important as a cornerstone for numerous application domains, such as human-computer interaction [11, 23, 24], augmented/virtual reality [21, 29], driver state monitoring [1, 16, 22]. Driven by deep learning, appearance-based gaze estimation methods have achieved significant progress [4, 5, 7, 37, 39]. When these pre-trained models are applied to new users, different background environments, or varying lighting conditions, their performance degrades sharply. Addressing this performance gap in gaze estimation has become a central objective and a recognized challenge in the field.

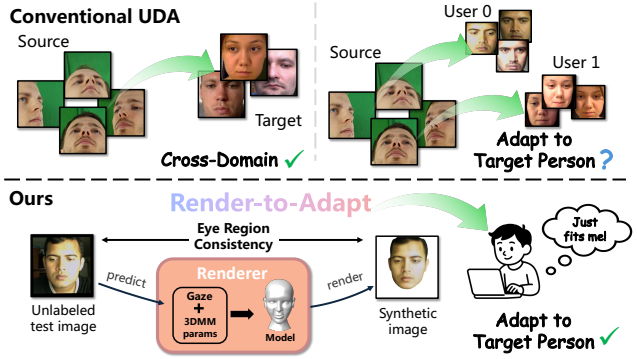


Figure 1. Comparison between conventional UDA methods and R2A. Conventional UDA methods aim to address general cross-domain discrepancies rather than adapting gaze models to specific individuals. In contrast, Render-to-Adapt (R2A) is designed for unsupervised personal adaptation, refining gaze estimation by enforcing eye-region consistency based on rendered data generated from predicted gaze.

To address this challenge, the research community has widely adopted Unsupervised Domain Adaptation (UDA) [2, 17, 28]. UDA methods assume access to a labeled source domain and an unlabeled target domain, and they improve the model's average performance on the target domain by aligning the data distributions of the two populations.However, in practical applications such as consumer electronics, the need for adaptation occurs at the individual-level, not the population-level assumed by UDA. The practical concern is not the model's average error across a population, but its performance and reliability on the specific individual currently using the system. UDA's pursuit of average performance improvement often comes at the cost of performance for some outlier individuals, which is unacceptable in real-world applications.

Therefore, the research focus should shift to a more practical new paradigm: Unsupervised Personal Adaptation (UPA). The task is defined as: given a pre-trained general-purpose model and a few unlabeled images from a single

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new user, how can we quickly and robustly adjust the model to minimize the gaze error for this specific user?

The core dilemma of the UPA task is: when faced with a single new user and no image labels, there is a lack of individual data for geometric calibration. Therefore, previous work [32] had to rely on auxiliary self-supervised tasks that are weakly correlated with the task's geometry. However, the recent emergence of high-fidelity differentiable rendering techniques such as GazeNeRF [20] and GazeGaussian [30] provides us with the ability to generate geometry-consistent personal data for new users, offering a brand new opportunity to solve this bottleneck.

Based on this, we will use a differentiable renderer as a tool to generate data for new users on the fly, and propose Render-to-Adapt (R2A), a self-supervised framework designed specifically for the UPA task. Our core insight is: if the gaze estimation model's initial gaze prediction for a real image is correct, then feeding the predicted gaze and this real image into the renderer to synthesize a new image, the key gaze cues extracted by the model from the real image and the synthetic image, namely the iris position, should be self-consistent. This Render-Cycle Consistency, as a self-supervised error signal, will directly calibrate the initial gaze prediction bias when backpropagated through the differentiable renderer. The main contributions of this work are as follows:

- This paper motivates the necessity of shifting from the UDA paradigm to the UPA paradigm, and experimentally demonstrates the systematic failure of SOTA UDA mechanisms on the UPA task.
- This paper proposes R2A framework, a novel self-supervised framework that utilizes a differentiable renderer and Render-Cycle Consistency to solve the problem of unlabeled personalized gaze adaptation.
- Under the Unsupervised Personal Adaptation paradigm, our R2A framework significantly outperforms existing SOTA methods on multiple cross-dataset person-specific optimization tasks, and robustly improves performance for all individual users.

2. Related Works

2.1. Gaze Estimation

Appearance-based gaze estimation has emerged as a prominent research topic[4, 5, 7, 37, 39]. However, these methods face a critical challenge: their performance degrades significantly when trained and tested across different domains, such as those varying in subjects, image quality, background environments, or lighting conditions. As a result, recent research has begun to explore cross-domain gaze estimation methods.

According to the availability of the source and target data, we review cross-domain gaze estimation methods under three settings. Domain Generalization: The target domain is unknown. Researchers have proposed various solutions, such as adversarial learning[6], data perturbation[33], feature separation[15], and multimodal fusion[34, 35], to improve the method's generalization ability.Unsupervised Domain Adaptation with source data: These methods utilize labeled source data and unlabeled target data, adapting the model to the target domain using techniques such as adversarial learning[26], outlier guidance[18], and contrastive regression[3].Unsupervised Domain Adaptation without source data: In the absence of true gaze supervision, researchers perform source-free adaptation by reducing uncertainty[2] , leveraging rotation consistency [28], or using outlier guidance[17].

However, in real-world deployment scenarios, the system only faces a single new user at any given time. This evaluation protocol is misaligned with real-world application needs. The aforementioned cross-dataset domain adaptation methods, while performing well on academic benchmarks, are misaligned with the real-world deployment scenarios of gaze estimation.

2.2. Personalized Gaze Estimation

We can divide personalized gaze estimation into two categories: Supervised and Few-Shot Personalization and Unsupervised Personal Adaptation.

Supervised and Few-Shot Personalization: Such methods rely on an extremely small number of labeled calibration points. Their core challenge is to overcome the overfitting of deep models on sparse data. The main approaches include the meta-learning paradigm, model offset predictors, or efficient on-device algorithms[9, 14, 19].

Unsupervised Personal Adaptation: These methods utilize only unlabeled images or video streams. Due to the lack of ground-truth supervision, they must rely on a surrogate signal. Mainstream techniques include Test-Time Adaptation driven by self-supervised tasks[31, 32] and representation learning based on gaze redirection[36].

3. Motivation

The motivation analysis in this section aims to elucidate why the prevailing paradigm of Unsupervised Domain Adaptation (UDA) is insufficient for addressing real-world gaze estimation challenges. We argue for a necessary shift towards a novel paradigm: Unsupervised Personal Adaptation (UPA). We seek to answer the following key questions through a series of validation experiments.

Question 1: Is the existing cross-domain gaze estimation paradigm, which aims to improve population-level average performance, practical in real-world scenarios?

Question 2: For this newly defined and more practical UPA task, are the existing UDA mechanisms still effective?

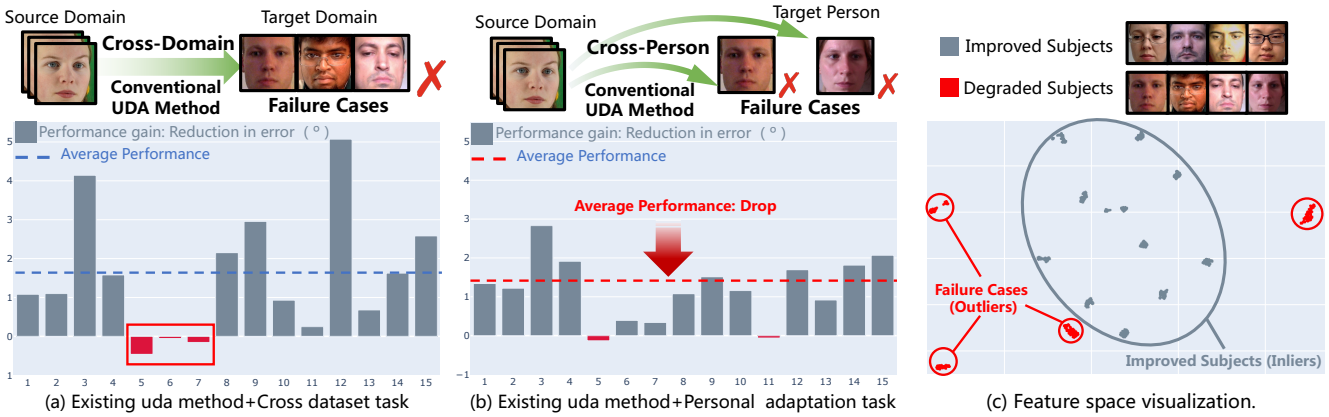


Figure 2. Limitations of conventional Unsupervised Domain Adaptation (UDA) mechanisms, motivating the shift to the Unsupervised Personal Adaptation (UPA) paradigm.

3.1. Dual Flaws of the Population UDA Paradigm

We first argue that the mainstream UDA paradigm suffers from two fundamental deficiencies.

First, its task setup is impractical. In real-world deployment scenarios, such as consumer electronics or driver monitoring, the system serves a single new user at a time. The core task is to adapt the model to that specific individual. However, the UDA paradigm mandates first constructing a target dataset comprising this user and dozens of others, then running a population-level adaptation, and finally applying this group-optimized model back to the original user. This task setup is severely misaligned with real-world application requirements.

Second, its performance is unreliable at the individual level. We executed a recent SOTA UDA method, UnReGA [2], pre-trained on the source domain $\mathcal{D}_E$ [40] and evaluated its cross-domain adaptation on the target domain $\mathcal{D}_M$ [38]. We analyzed the error change for each subject post-adaptation. As highlighted by the red boxes in Fig. 2 (a), a subset of subjects experienced an increase in error post-adaptation. This reveals that UDA's pursuit of improved population-level averages comes at the cost of performance stability and reliability for individual users.

Answer to Question 1: The UDA paradigm is impractical in its task setup and unreliable in its outcomes. We posit that the field's focus must shift to a more rational and practical paradigm: Unsupervised Personal Adaptation (UPA). The goal of UPA is to rapidly calibrate a pre-trained model using only a small number of unlabeled images from a single new user.

3.2. Limitations of UDA Mechanisms for the Personal Adaptation Task

Given the UDA paradigm's impracticality, we have proposed the more rational UPA task. A natural question thus arises: Can the existing UDA mechanisms (the algorithmic mechanisms) be repurposed for this new individual optimization task?

To answer this question, we apply SOTA UDA algorithms to the UPA task, which involves running adaptation for each individual separately (Source $\rightarrow \{\text{User}_i\}$).

Fig. 2 (b) illustrates the performance of UDA mechanisms when applied to this individual-level task. The bar chart indicates that many subjects (red bars) exhibit increased error post-adaptation, confirming that individual-level performance remains unreliable. To investigate which individuals fail in this setup, we extracted high-dimensional facial appearance feature vectors for all subjects and visualized them using UMAP. Fig. 2 (c) clearly reveals that the failure cases (red dots) are almost exclusively distributed outside the 80% confidence ellipse, identifying them as outliers in the feature space.

This finding directly answers Question 2: UDA mechanisms are, by design, ill-suited for the single-user optimization task. They tend to pull the model towards the population mean, which perversely harms the performance of the very outliers who require adaptation the most.

3.3. The Necessity of a New Paradigm

Our motivation analysis provides a clear two-fold conclusion: The existing UDA paradigm is impractical for real-world applications due to its task setup; existing UDA mechanisms are algorithmically unsuitable for solving the new UPA task.

Therefore, to address real-world applications, it is imperative to develop entirely new techniques specifically for the more practical UPA paradigm—techniques that do *not* rely on population-level statistics. This provides a strong motivation for the novel method we introduce in Sec. 4.

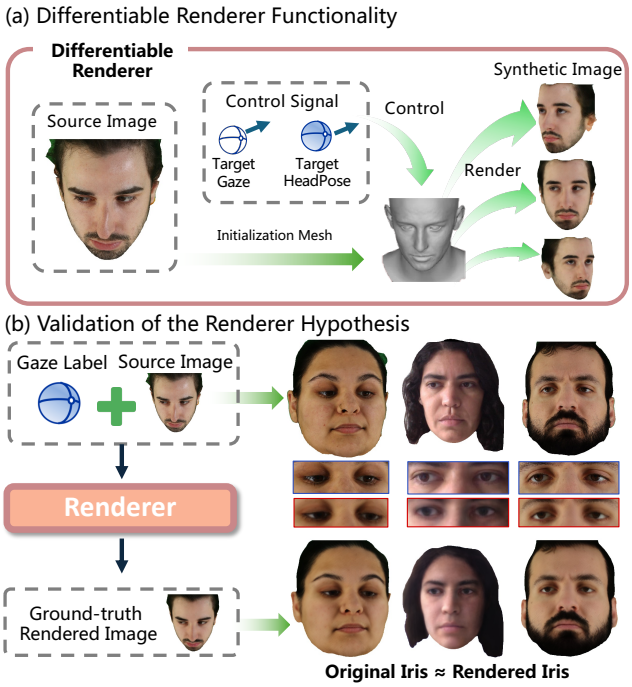


Figure 3. Differentiable Renderer Functionality and Hypothesis Validation. (a) Function: The renderer R uses the source image I_{real} as an appearance template and accepts the target gaze g_{target} and head pose p_{target} as control signals to synthesize diverse new images. (b) Hypothesis Validation: We verify that R possesses high gaze-to-iris fidelity. When the ground-truth gaze label g_{gt} is used as the control signal, the iris position in its synthesized image I_{recon} is visually almost identical to the iris position in the original image I_{real}.

4. Method

4.1. Framework Overview and Formulation

Our method aims to solve the Unsupervised Personal Adaptation (UPA) task proposed in the Motivation section. Given a pre-trained general-purpose multi-task gaze estimator E_θ and a small, unlabeled calibration set $\mathcal{I}_U = \{I_{real}\}$ from a single new user, the goal is to generate a personalized model E_{θ^*} that is optimized for this user.

As shown in Fig. 4, we propose the Render-to-Adapt (R2A) framework. Its core principle is to utilize Render-Cycle Consistency as a self-supervised signal. We hypothesize that if a model's gaze prediction g_{pred} for an image I_{real} is accurate, then after feeding this g_{pred} into a differentiable renderer R to synthesize a new image I_{recon}, the model's predictions for the key gaze cue (iris position) should remain consistent between I_{real} and I_{recon}.

We verified the rationality of this hypothesis through visualization, as shown in Fig. 3 (b): when using the ground-truth gaze label g_{gt} as input to the renderer R, the generated I_{recon} exhibits high fidelity in iris position compared to I_{real}. This confirms that R can accurately translate changes in the gaze vector into corresponding changes in the iris position, providing a solid foundation for our self-supervised signal.

The R2A workflow is an Analyze-Synthesize-Re-analyze loop.Analyze: E_θ analyzes I_{real}, predicting g_{pred} and the iris position $iris_{pred_real}$. Synthesize: The differentiable renderer R uses g_{pred} to synthesize a new image $I_{recon} = R(I_{real}, g_{pred})$. Re-analyze: E_θ re-analyzes I_{recon}, predicting $iris_{pred_recon}$.

When $iris_{pred_real} \neq iris_{pred_recon}$, the resulting consistency loss $\mathcal{L}_{\text{cycle}}$ (detailed in Sec. 4.4.2) is backpropagated through the differentiable renderer R, thereby unsupervisedly calibrating the initial gaze prediction bias of E_θ.

4.2. Core Architectural Components

4.2.1. Pre-trained Multi-task Estimator

E_θ is our target for calibration. It is a deep neural network whose parameters θ have been pre-trained on large-scale gaze datasets. The network is designed as a multi-task model:

$$(g_{\text{pred}}, iris_{\text{pred}}) = E_\theta(I; \theta) \tag{1}$$

Where the input I is a facial image, and the outputs are a gaze vector $g_{\text{pred}} \in \mathbb{R}^2$ and iris contour points $iris_{\text{pred}}$ in 2D image coordinates. Specifically, $iris_{\text{pred}}$ ($\in \mathbb{R}^{k\times 17\times 2}$) comprises 17 (x,y) coordinate points for each of the $k = 2$ eyes.

4.2.2. Auxiliary Iris Prior

D_{aux} is a non-learning traditional computer vision module. We use it to extract iris position pseudo-labels $iris_{pseudo}$ from the real image I_{real}:

$$iris_{pseudo} = D_{\text{aux}}(I_{\text{real}}) \tag{2}$$

This module can detect facial landmarks via Dlib to locate the eye regions, and then use the Hough Circle Transform in OpenCV to estimate the iris center. Its role is not to provide perfect labels, but to anchor the iris detection sub-task of E_{θ_i}.

4.2.3. Fixed-Parameter Differentiable Renderer

The key to our method is R, a differentiable renderer with frozen parameters ϕ, capable of rendering a facial image according to a given target gaze:

$$I_{\text{recon}} = R(I_{\text{source}}, g_{\text{target}}; \phi) \tag{3}$$

This function takes I_{source} as the source image to extract appearance information such as identity, texture, and lighting, and uses g_{target} as the target gaze vector to synthesize a reconstructed image I_{recon}. Critically, the head pose from I_{source} is preserved during this synthesis, so that only the gaze direction is altered in I_{recon}. R is differentiable with respect to its inputs (especially g_{target}). When a loss is computed on its output I_{recon}, the gradient of this loss with respect to R's input g_{target} can be calculated via the chain rule.

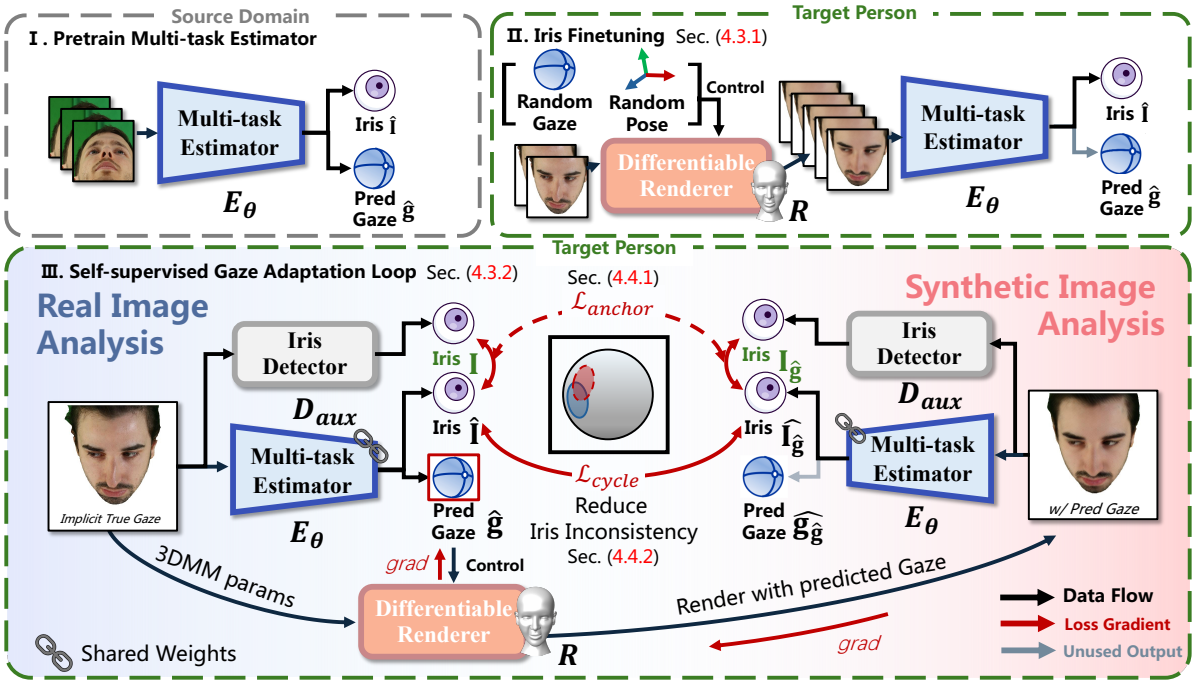


Figure 4. Overview of the R2A framework. (I) Pre-training: A multi-task estimator E_θ is first trained on the source domain. (II) Iris Finetuning: The renderer R generates diverse images for the target individual, and the auxiliary prior D_{aux} is used to finetune the iris head of E_θ. (III) Self-supervised Calibration: The core loop. E_θ predicts g_{pred}, and the renderer R synthesizes I_{recon} accordingly. The $\mathcal{L}_{cycle}$ loss calibrates the gaze head by comparing the iris prediction consistency between I_{real} and I_{recon}, while the $\mathcal{L}_{anchor}$ loss continues to anchor the iris head.

In our implementation, we employ the pre-trained weights of GazeGaussian [30] as the renderer R.

4.3. Calibration Flow

Our personalization calibration flow is divided into two sequential stages: (1) a pre-finetuning stage to strengthen the iris predictor; and (2) a self-supervised calibration loop to correct gaze bias.

4.3.1. Preliminary Iris Task Finetuning

The goal of this stage is to strengthen the iris prediction capability of E_θ before the core gaze calibration begins.

We first perform data augmentation: using the fixed differentiable renderer R, we combine the user's few real images $I_{real} \in \mathcal{I}_U$ with randomly generated gaze directions and head poses to synthesize diverse new images I_{synth}.

Next, we use the auxiliary iris detector D_{aux} to extract stable iris pseudo-labels for the augmented dataset (containing both I_{real} and I_{synth}). We use only the $\mathcal{L}_{\text{anchor}}$ loss (detailed in Sec. 4.4.1) to fine-tune the model's iris prediction head E_{θ_i}, aligning its predictions with the priors provided by D_{aux}.

4.3.2. Self-supervised Gaze Adaptation Loop

The goal of this stage is to use $\mathcal{L}_{\text{anchor}}$ and $\mathcal{L}_{\text{cycle}}$ jointly as a self-supervised signal to optimize the estimator E_θ and calibrate its gaze bias by iterating over the few unlabeled user images $\{I_{real}\}$. We perform the following operations:

1. **Sample Image:** Take an image I_{real} from $\mathcal{I}_U$.
2. **Execute Forward Pass (to obtain variables for loss):**
   (a) Prior Analysis: $iris_{pseudo_real} = D_{\text{aux}}(I_{real})$
   (b) Initial Analysis: $(g_{\text{pred}}, iris_{\text{pred_real}}) = E_\theta(I_{real})$
   (c) Synthesize: $I_{\text{recon}} = R(I_{real}, g_{pred})$
   (d) Re-analyze: $(_, iris_{\text{pred_recon}}) = E_\theta(I_{recon})$
   (e) Prior Analysis: $iris_{pseudo_recon} = D_{\text{aux}}(I_{recon})$
3. **Calculate Loss:** Use the items (a)-(e) generated in step 2 to compute the total loss defined in Sec. 4.4.3: $\mathcal{L}_{\text{total}} = \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor}} + \lambda_{\text{cycle}}\mathcal{L}_{\text{cycle}}$.
4. **Backpropagate:** Compute the gradient of $\mathcal{L}_{\text{total}}$ and update the parameters θ of the estimator E_θ.

By repeating this loop over $\mathcal{I}_U$, the gradient from the $\mathcal{L}_{\text{cycle}}$ loss (detailed in Sec. 4.4.2) will backpropagate through R, thereby iteratively correcting the bias in g_{pred}.

4.4. Optimization Objectives

The parameters θ of the estimator E_θ are fine-tuned by minimizing a total loss function, which is composed of two

novel components: a prior-based anchoring regularization loss $\mathcal{L}_{\text{anchor}}$ and a render-cycle keypoint consistency loss $\mathcal{L}_{\text{cycle}}$.

4.4.1. Prior-based Anchoring Loss

The primary role of this loss, $\mathcal{L}_{\text{anchor}}$, is to regularize the iris prediction head (E_{θ_i}). It utilizes the stable priors $iris_{pseudo_real}$ and $iris_{pseudo_recon}$, extracted by D_{aux} from I_{real} and I_{recon} respectively, as pseudo-labels to prevent E_θ from drifting or collapsing during the self-supervised learning process.

We apply this anchoring loss to both the real and reconstructed domains separately, supervising with the pseudo-labels from their respective domains to ensure E_{θ_i} remains stable across both:

$$\mathcal{L}_{\text{anchor}} = \mathcal{L}_1(iris_{\text{pred_real}}, iris_{\text{pseudo_real}}) + \mathcal{L}_1(iris_{\text{pred_recon}}, iris_{\text{pseudo_recon}}) \quad (4)$$

4.4.2. Render-Cycle Iris Consistency

$\mathcal{L}_{\text{cycle}}$ is the core of our framework and the main self-supervised signal that drives the adaptation of the gaze head E_{θ_g}:

$$\mathcal{L}_{\text{cycle}} = \mathcal{L}_1(iris_{\text{pred_real}}, iris_{\text{pred_recon}})$$

If the initial g_{pred} is incorrect, then I_{recon} will be a flawed reconstruction, with the rendered gaze being incorrect. This will produce an $iris_{pred_recon}$ that is different from the original $iris_{pred_real}$, leading to a non-zero $\mathcal{L}_{\text{cycle}}$. The gradient from this loss will backpropagate to correct g_{pred}.

4.4.3. Total Loss

Our final objective function for unsupervised test-time adaptation is a weighted sum of the prior regularization and the self-supervised consistency loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor}} + \lambda_{\text{cycle}}\mathcal{L}_{\text{cycle}} \quad (5)$$

Empirically, we set $\lambda_{\text{anchor}} = \lambda_{\text{cycle}} = 1$.

5. Experiments

As Unsupervised Personal Adaptation is an emerging paradigm, it lacks dedicated datasets and experimental protocols. Building upon the experimental setup of [32], we re-purpose existing gaze datasets and design more extensive experiments to investigate the R2A framework's effectiveness, its individual components, and its personal calibration performance.

5.1. Data Preparation

To rigorously evaluate model performance on the UPA task, we repurposed five widely-used gaze estimation benchmark datasets: ETH-XGaze ($\mathcal{D}_E$) [40], Gaze360 ($\mathcal{D}_G$) [12], GazeCapture ($\mathcal{D}_C$) [13], MPIIGaze ($\mathcal{D}_M$) [38], and EyeDiap ($\mathcal{D}_D$) [8]. Among them, **ETH-XGaze** is collected in a laboratory environment using 18 SLR cameras, containing 756k high-quality images from 80 subjects. **Gaze360** contains 101k images captured on streets using a 360° camera. **GazeCapture** is collected from 150 subjects using mobile phones and tablets in daily environments, from which we use 179k images. **EyeDiap** contains 16k images from 16 users, captured in a laboratory setting with screens and 3D floating balls. **MPIIGaze** is captured by web cameras from 15 subjects during daily laptop use, with each subject providing 3,000 images.

Our UPA experimental protocol treats each subject in the target dataset as an independent, label-free adaptation task. In our experiments, we use $\mathcal{D}_E$ or $\mathcal{D}_G$ as the labeled source data, and then fine-tune and test the model on each individual in the target datasets $\mathcal{D}_M$, $\mathcal{D}_D$, and $\mathcal{D}_C$, respectively. This cross-dataset protocol models diverse, real-world scenarios for individual-level generalization across different data sources.

5.2. Implementation Details

We implement our method using PyTorch. The estimator E_θ uses a ResNet-based architecture [10] as a shared backbone. This backbone is followed by two output heads: (a) The Gaze Head, which outputs a 2D vector (yaw, pitch); and (b) The Iris Head, which outputs a 68D vector representing the 34 2D contour point coordinates for both eyes. We pre-train this model for 10 epochs using the Adam optimizer with a learning rate of 10^{-4}. The differentiable renderer (R) utilizes pre-trained weights from [20, 30].

5.3. Effectiveness of R2A Framework

As shown in Tab. 1, we establish four adaptation strategies as comparative baselines to clearly demonstrate the performance of our R2A.

Without adaptation: This represents the baseline model tested directly on the target individual. The results show the highest error, exposing the unresolved personalization gap. Unsupervised Domain Adaptation: Represented by UnReGA. This strategy aligns the group-level distribution and significantly outperforms the baseline, but it is not optimized for individuals. Supervised Personal Adaptation: This method fine-tunes the model using the individual's labels. It serves as the performance upper bound for our unsupervised UPA task and achieves the lowest error.

Compared to these baselines, our R2A demonstrates significant performance improvements. First, R2A substantially outperforms both the Without adaptation baseline and the group-level UDA strategy across all cross-dataset settings.Second, our Unsupervised Personal Adaptation method efficiently narrows the performance gap to Supervised Personal Adaptation. Notably, on the $\mathcal{D}_G\rightarrow$

Table 1. Quantitative comparison of adaptation strategies on cross-dataset personalization benchmarks.

Adaptation Strategy	**Method**	$\mathcal{D}_E \rightarrow \mathcal{D}_M$	$\mathcal{D}_E \rightarrow \mathcal{D}_D$	$\mathcal{D}_E \rightarrow \mathcal{D}_C$	$\mathcal{D}_G \rightarrow \mathcal{D}_M$	$\mathcal{D}_G \rightarrow \mathcal{D}_D$	$\mathcal{D}_G \rightarrow \mathcal{D}_C$
Without adaptation	Baseline	8.75	9.08	8.86	10.13	9.58	9.49
Unsupervised Domain Adaptation	UnReGA[2]	5.11 ▼ 3.64°	5.70 ▼ 3.383°	5.75 ▼ 3.11°	5.42 ▼ 4.71°	5.80 ▼ 3.78°	6.52 ▼ 2.97°
Supervised Personal Adaptation	Baseline + Supervised FT	4.30 ▼ 4.45°	4.12 ▼ 4.97°	4.75 ▼ 4.11°	5.38 ▼ 4.74°	4.96 ▼ 4.62°	5.28 ▼ 4.21°
Unsupervised Personal Adaptation	R2A(Ours)	4.84 ▼ 3.91°	5.28 ▼ 3.80°	5.06 ▼ 3.80°	4.92 ▼ 5.21°	5.43 ▼ 4.15°	5.67 ▼ 3.82°

Table 2. Quantitative comparison with state-of-the-art methods, categorized by their adaptation strategy.$*$ indicates methods originally designed for group-level UDA, which we reproduced and adapted to our UPA task.

Method	$\mathcal{D}_E \rightarrow \mathcal{D}_M$	$\mathcal{D}_E \rightarrow \mathcal{D}_D$	$\mathcal{D}_G \rightarrow \mathcal{D}_M$	$\mathcal{D}_G \rightarrow \mathcal{D}_D$
Setting: Without adaptation				
PureGaze[6]	7.08	7.48	9.28	9.32
CDG[27]	6.73	7.95	7.03	7.27
Xu et al.[33]	6.50	7.44	7.55	9.03
Liang et al.[15]	5.79	6.96	7.06	7.99
CLIP-Gaze[35]	6.41	7.51	6.89	7.06
LG-Gaze[34]	6.45	7.22	6.83	6.86
Setting: Unsupervised Domain Adaptation				
Gaze360[12]	5.97	7.84	7.38	9.61
GazeAdv[25]	6.75	8.10	8.19	12.27
PnP-GA[17]	5.53	5.87	6.18	7.92
CRGA[28]	5.68	5.72	6.09	6.68
UnReGA[2]	5.11	5.70	5.42	5.80
Setting: Unsupervised Personal Adaptation				
PnP-GA*	6.39	7.27	6.57	7.66
UnReGA*	5.72	6.07	5.95	6.27
ELF-UA[32]	6.79	-	7.38	-
R2A(Ours)	**4.84**	**5.28**	**4.92**	**5.43**

$\mathcal{D}_M$ task, our method (4.92°) even surpasses the supervised fine-tuning (5.38°).These results confirm the effectiveness of R2A and validate that shifting to the individual-level UPA task is a more effective and precise solution than adhering to the group-level UDA paradigm.

5.4. Comparison with SOTA Methods

To demonstrate the superiority of the R2A framework, we present a comprehensive comparison in Tab. 2 against SOTA methods under various adaptation strategies, clearly illustrating the SOTA performance within each paradigm. For a fair comparison, we follow our proposed UPA task setting. We reproduce the mechanisms of SOTA UDA methods (PnP-GA and UnReGA) and adapt them to our UPA task, which we denote as '$*$'.

The results confirm that UDA mechanisms designed to align group-level statistics do not generalize well to the individual optimization task, corroborating our findings in Sec. 3. In stark contrast, our R2A is specifically designed for the UPA task. It achieves SOTA performance across all benchmarks, significantly outperforming all other unsupervised baselines.

Table 3. Ablation study of the R2A framework. We analyze the contribution of each component, the impact of sample size, and the effect of different loss functions.

Method	$\mathcal{D}_E \rightarrow \mathcal{D}_M$	$\mathcal{D}_E \rightarrow \mathcal{D}_D$	$\mathcal{D}_G \rightarrow \mathcal{D}_M$	$\mathcal{D}_G \rightarrow \mathcal{D}_D$
ResNet-18	8.75	9.08	10.13	9.58
ResNet-18 + oma	7.23	7.49	7.69	8.50
ResNet-18 + oma + js	6.29	6.45	6.24	6.17
ResNet-18 + oma + js + sg	4.84	5.28	4.92	5.43
ResNet-50	7.10	7.47	8.21	9.34
ResNet-50 + oma	6.22	6.39	7.25	8.55
ResNet-50 + oma + js	5.79	6.14	5.8	6.25
ResNet-50 + oma + js + sg	4.64	5.12	5.01	5.28
ResNet-18+oma+js+sg:diff sample num				
num=1	5.78	6.11	6.03	6.05
num=5	4.99	5.48	5.36	5.68
num=10	4.84	5.28	4.92	5.43
num=20	4.77	5.20	4.89	5.12
ResNet-18+oma+js+sg:differentiable model				
GazeNeRF[20]	5.13	5.69	5.44	6.05
GazeGaussian[30]	4.84	5.28	4.92	5.43
$\mathcal{L}_{\text{anchor}}$	6.29	6.45	6.24	6.17
$\mathcal{L}_{\text{cycle}}$	5.21	5.86	5.43	6.07
$\mathcal{L}_{\text{anchor}}$ + $\mathcal{L}_{\text{cycle}}$	4.84	5.28	4.92	5.43

5.5. Ablation Study

To validate the effectiveness and necessity of each component in our framework, we conducted a series of detailed ablation studies, with the results shown in Tab. 3.

Component-wise Analysis, we analyze the composition of the R2A framework by progressively adding components.

- Baseline (Source Model): The baseline model exhibits the highest error when applied directly to the target individual.
- + oma: Adding the iris task during pre-training guides the model to focus on key geometric features, leading to an initial reduction in error.
- + js (Stage 1: Iris Pre-finetuning): Through Stage 1 fine-tuning, the model further adapts to the individual's unique appearance, causing the error to continue to decrease.
- + sg (Stage 2: Self-supervised Calibration): Introducing the Stage 2 self-supervised loop (i.e., the complete R2A framework) brings the most significant performance improvement, substantially calibrating the error to the lowest level.

Impact of Sample Size: We analyze the effect of the number of samples used in Stage 2 (sg). Using only 1 image achieves the most significant error calibration, drastically

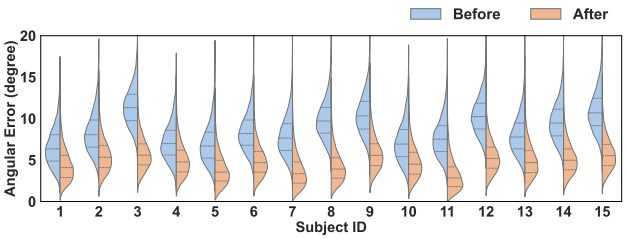


Figure 5. Error distribution comparison before and after adaptation. R2A not only shifts the main error body downwards but also significantly suppresses extreme outliers, enhancing model reliability.

reducing the error from the high baseline level. When the number of samples increases to 10, the model performance tends to saturate, achieving the best balance between efficiency and accuracy. Continuing to increase to 20 images results in only marginal performance improvement. This demonstrates that R2A can achieve robust personalized calibration with only a very small number of unlabeled images.

Analysis of Loss Functions: The ablation data clearly demonstrates the synergistic effect of our two core loss functions, $\mathcal{L}_{\text{anchor}}$ and $\mathcal{L}_{\text{cycle}}$. $\mathcal{L}_{\text{anchor}}$ is responsible for preventing the model from drifting during the self-supervised process, while $\mathcal{L}_{\text{cycle}}$ performs precise gaze correction on this stable foundation. The combination of the two achieves the optimal performance.

5.6. Analysis of Personal Calibration Performance

To comprehensively analyze the effectiveness of the R2A framework at the individual level, we conduct an in-depth, subject-by-subject analysis of the performance on the $\mathcal{D}_E \rightarrow \mathcal{D}_M$ task.

Error Distribution: Fig. 5 provides an in-depth look at the complete error distribution before and after calibration. The results show that R2A achieves key objectives: not only does it shift the main body of the entire error distribution to a lower error range, which reflects a clear improvement in average accuracy; it also significantly shortens the tail of the distribution, greatly suppressing outliers. This proves that R2A not only improves average accuracy but, more critically, enhances the system's stability and reliability.

Qualitative Results: Fig. 6 provides qualitative visual examples. These images intuitively confirm our quantitative results: R2A is capable of generating accurate gaze predictions that are highly consistent with the ground truth labels through self-supervised calibration.

6. Conclusion

This paper first demonstrates the shortcomings of the existing Unsupervised Domain Adaptation paradigm, which is impractical in its task setup and unreliable in its individual performance, and advocates that the research focus should shift to the more practical Unsupervised Personal Adaptation paradigm. We propose Render-to-Adapt (R2A), a novel self-supervised framework designed specifically for the UPA task. It creates an individual-specific, self-supervised signal based on the Render-Cycle Consistency of key gaze cues, which enables quick calibration of a pre-trained model to adapt to a single new user without requiring any labels. We demonstrate through experiments that R2A not only significantly outperforms "repurposed" SOTA UDA mechanisms but also surpasses other UPA methods.

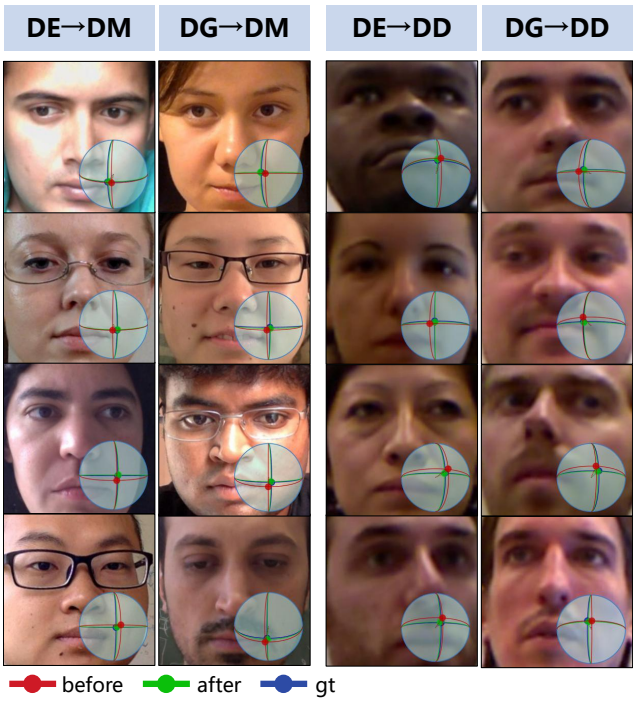


Figure 6. Qualitative visualization results. The baseline predictions exhibit clear deviations, whereas R2A, through self-supervised calibration, generates accurate gaze estimations that are visually highly consistent with the ground truth labels.

Limitations and Future Work. The performance of R2A relies on the differentiable renderer and the iris detector. In extreme scenarios with harsh lighting or heavy occlusions, if these components fail to accurately reconstruct the user's unique eye features or locate the iris, the quality of the self-supervised signal may degrade, affecting calibration performance. Additionally, while the personalization process is efficient—achieving saturation with only 5 images in approximately 2 seconds (on RTX 4090) and introducing zero overhead during inference—the initial optimization still requires extra computational resources compared to direct inference. Future work could explore extending the Render-Cycle Consistency to other personalized tasks with strong geometric constraints, such as head pose estimation, or developing more lightweight rendering modules to further reduce the adaptation latency.

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